

Production of *Arthrospira platensis* in open ponds

Correction

Partie 1 : Evaporation

Question 1

There are two characteristic lengths (L and l), hence two Reynolds numbers:

$$Re_L = \frac{v_{wind} L}{\vartheta_{air}} = 16.9 \times 10^6$$
$$Re_l = \frac{v_{wind} l}{\vartheta_{air}} = 3.39 \times 10^6$$

Question 2

A correlation must be chosen. The criteria for choosing one are the following:

- configuration: flat plate / tube? => flat plate
- flow regime: laminar / turbulent / both (one then the other) => both

We will therefore use $Sh_L = (0.037 Re_L^{4/5} - 871) Sc^{1/3}$ after checking the range of validity.

The Reynolds numbers are obviously correct. The Schmidt number of water in air must be calculated:

$$Sc_{water/air} = \frac{\vartheta_{air}}{D_{water/air}} = 0.651$$

The second criterion is valid ($0.6 < Sc < 60$), so we can use the correlation and obtain for both characteristic lengths:

$$Sh_L = \frac{k_L L}{D_{water/air}} = 18.7 \times 10^3$$
$$Sh_l = \frac{k_l l}{D_{water/air}} = 4.61 \times 10^3$$

Question 3

To calculate the evaporation rates (Ψ) we need the mass transfer coefficients available to us (Question 2), the surface area of the pond ($A_{pond} = l \times L = 500 \text{ m}^2$) and the transfer potential, i.e. the concentration difference.

At the surface of the pond, the partial pressure of water is the vapour pressure of water at 35 °C: $P^{sat}(35 \text{ °C}) = 5626.7 \text{ Pa}$. Using the law of perfect gases, it is possible to obtain the concentration of water at this point:

$$C_{sat} = \frac{P^{sat} M_{water}}{RT} = 0.0395 \text{ kg/m}^3$$

Away from the raceway, the relative humidity is $Y_{\infty} = 20\%$, which means that the concentration is :

$$C_{\infty} = Y_{\infty} C_{sat} = 7.91 \cdot 10^{-3} \text{ kg/m}^3 \text{ (assuming the air is also } 35^{\circ}\text{C)}$$

Then we can calculate the evaporation rates for the two characteristic lengths:

$$\begin{aligned}\Psi_L &= A_{pond} k_L (C_{sat} - C_{\infty}) = 0.149 \text{ kg/s} \\ \Psi_l &= A_{pond} k_l (C_{sat} - C_{\infty}) = 0.184 \text{ kg/s}\end{aligned}$$

Question 4

Using an average evaporation rate between Ψ_L and Ψ_l , we get the amount of water that is lost over the 10 days of cultivation :

$$V_{evap} = \frac{\Psi_L + \Psi_l}{2 \rho_{water}} \times \frac{8 \text{ h}}{\text{jour}} \times 3600 \frac{\text{s}}{\text{h}} \times 10 \text{ jours} = 47.9 \text{ m}^3$$

which corresponds to a height of $\delta = \frac{V_{evap}}{A_{pond}} = 9.59 \text{ cm}$ to be compared to the initial 30 cm of the pond

Question 5

The diffusion rate of water through a polypropylene sheet of $e_{sheet} = 200 \text{ }\mu\text{m}$ can be estimated from Fick's law :

$$\Psi_{Diff} = -D_{WaterPP} \nabla C \cdot A_{Pond} \approx -D_{Water/PP} \frac{C_{\infty} - C_{sat}}{e_{sheet}} \cdot A_{Pond} = 1.58 \cdot 10^{-8} \text{ kg/s}$$

Integrated over 10 days (for 24 hours), the total volume is 14 ml, which is negligible compared to the total volume of the pond. Furthermore, potential leaks from the construction of the pond would be greater than this volume.

Part 2 : CO₂ supply

Question 6

Laplace's law is used to obtain the overpressure in the bubble and Henry's law to obtain the concentration.

$$P_{bulle} = 1 \text{ atm} + \Delta P$$

CO₂ represents 2% of this pressure. The concentration at the interface is therefore:

$$C_{CO_2, interface} = H_{CO_2/water} \times P_{CO_2, bulle} = H_{CO_2/water} \times 0.02 \times P_{bulle}$$

Which can then be converted to mg/L

Question 7

Simply use the correlation $Sh = 1.35 Re^{0.5} Sc^{0.5}$

It can be commented that the large bubble has a higher Sherwood, but don't be fooled, its transfer coefficient is lower.

Question 8

Multiply the transfer coefficient by the surface area of the bubble and the exchange potential:

$$\psi = k_l \pi d_{bulle}^2 (C_{CO2,interface} - C_{\infty})$$

With $C_{\infty} = 11 \text{ mg/L}$

It can be commented that the large bubble has a larger exchange surface area which compensates for the k_l and allows it to transfer more gas.

Question 9

To get the number of bubbles per unit time, divide the flow rate by the volume of a bubble.

$$N_{bulles} = 6 Q / (\pi d_{bulle}^3)$$

The number of small bubbles is much greater than the number of large bubbles.

Question 10

The total flow is the product of the number of bubbles and the flow of a bubble

$$\psi_{Total} = \psi \times N_{bulles}$$

In the end, the bubbler offering small bubbles is 33 times more efficient.

Type de bulle	Petite	Grande	Unité
Dbulle	1.00E-04	1.00E-03	m
ΔP	2884	288.4	Pa
Pbulle	104209	101613.4	Pa
$C_{CO2,interface}$	28.1	27.4	mg/L
Re	56	563	-
Sh	260	823	-
k_l	4.14E-03	1.31E-03	m ³ /s
ψ	2.22E-06	6.73E-05	mg/s
Nbulles	3.18E+09	3.18E+06	Bulles
ψ_{Total}	7060	214	mg/s